

**AMENDMENT NO. 2 MARCH 2007
TO
IS 6582 (PART 2) : 2001 BIO-ASSAY METHOD FOR
EVALUATING ACUTE TOXICITY OF INDUSTRIAL
EFFLUENTS AND WASTE WATERS**

**PART 2 USING TOXICITY FACTOR TO ZEBRA FISH
(BRACHYDANIO RERIO)**

(First Revision)

(Page 1, clause 5) — Substitute the following for the existing text:

5 SAMPLING

5.1 Point of Sampling

In those cases where the effluent at a specific point is to be tested, the question of choosing the point of sampling does not arise. However, where the composition of an effluent as finally discharged by the sewage treatment plant is to be ascertained, the point of sampling shall be the final outlet of the treatment plant.

5.2 Frequency of Sampling

When it is required to find out variations in the composition of the effluent during a specified period, such as that of peak discharge, the samples shall be taken at short and appropriate intervals, say, every 5, 10, 15 or 30 min and analysed. To study the average conditions over a cycle of operations or a period (usually 24 hours) or during the daily working period of the treatment plant, the collection of composite sample shall be adopted. The composite sample shall be made by collecting at appropriate intervals samples from the common channel or drain at a point where the flow of the effluent is likely to be most representative of the entire volume, and mixing. The volume of the individual samples shall be a fixed proportion of the volume of the effluent flowing at that time. The interval should depend upon the frequency of variation in the nature of the effluent and the volume of flow. Care shall be taken to take the samples in such a way as to maintain the true proportion of suspended solids. Samples shall not be taken by skimming the top or scraping the bottom. A point about one-third of the way from the bottom shall normally be selected. The samples

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shall be drawn gently without unnecessary aeration. In most cases, collection of samples every hour would be sufficient.

5.3 Sampling Instrument

Porcelain-lined or enamelled pails, in which the lining is unbroken, or glass vessels shall be used for taking samples. The vessels used for taking the sample shall be wide-mouthed and small enough for the contents to be transferred quickly to the sample container without leaving behind any deposit or scum. Automatic sampling devices, if available, may be used.

5.3.1 Each individual sample shall be deposited in a receptacle of sufficient size to hold the entire composite sample. Clean and dry carboys, other large glass containers or enamelled buckets with lids without chipping may be used for pooling the sample.

5.4 Sample Containers

5.4.1 The quantity of sample required for analysis shall be taken from the composite sample after thorough mixing in order to keep the solids in suspension.

5.4.2 The sample for analysis shall be drawn in clean glass-stoppered bottles, which shall be rinsed with a portion of the sample. New bottles shall be washed with acid and thoroughly rinsed with distilled water before being brought into use. About 2 to 3 litres of the sample will be required for analysis. The bottle containing the final sample shall be filled so that a small air bubble is present after closure to prevent leakage or even breakage arising from any subsequent changes in temperature. The stopper shall be firmly inserted and, if the sample is to be transported some distance, tied down to keep in position.

5.4.3 The label on the bottle shall bear the name of the sampling authority, details of the type of sample, place, date and time of sampling.

5.5 Preservation of Samples

5.5.1 The samples shall be kept at a low temperature (about 4°C) during collection and thereafter.

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5.5.2 No single method of preservation is applicable for the sample for all the tests. The analysis shall be carried out, preferably, immediately after collection. Storage at 3° to 4°C in a well insulated ice box or refrigerator is the best way to preserve most samples till the next day. Where chemical preservatives are used as specified for individual tests, these shall be added to each portion of sample taken for the particular test and not to the entire sample.

(CHD 32)

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